Gaussian distributions, where we have invariant (stationary) averages

$$\langle q \rangle = \langle p \rangle = 0,$$

and

$$\langle \Omega^2 q^2 \rangle = \langle p^2/m \rangle = k_{\mathrm{B}}\mathrm{T}, \qquad \langle qp \rangle = 0.$$

We expect that the discretization schemes that we employ will introduce errors in the computed values of the invariant (long-time) averages. In the large-time limit, these averages will converge towards their invariant value, which we can find using the procedure outlined in [59].

For the harmonic oscillator, the force $-U'(q)$ will be linear, and therefore we would expect that any update step in an algorithm will be linear in the state variables q and p. Thus, denoting the iteration index as a subscript, we may explicitly write one iteration of a general numerical method for evolving the dynamics as

$$\begin{bmatrix} q_{n+1} \\ p_{n+1} \end{bmatrix} = \Psi \begin{bmatrix} q_n \\ p_n \end{bmatrix} + \mu_n, \tag{7.7}$$

where Ψ is a constant matrix that depends on the stepsize h, the friction coefficient γ, the particle mass m and the spring constant Ω. Additionally μ_n is a vector of stochastic processes. For example, in the BBK method we can compute directly from its update scheme that

$$\Psi^{\mathrm{BBK}} = \begin{bmatrix} 1 - \frac{h^2\Omega^2}{2m} & \frac{h}{m}\left(1 - \frac{\gamma h}{2}\right) \\ \frac{h\Omega^2}{2m}\left(\frac{h^2\Omega^2 - 4m}{2+\gamma h}\right) & \frac{(2-\gamma h)\left(2m - h^2\Omega^2\right)}{2m(2+\gamma h)} \end{bmatrix},$$

$$\mu_n^{\mathrm{BBK}} = \begin{bmatrix} \sqrt{\frac{k_{\mathrm{B}}\mathrm{T}\gamma h^3}{2m}}\mathrm{R}_n \\ \frac{\sqrt{2k_{\mathrm{B}}\mathrm{T}\gamma hm}}{2+\gamma h}\left(\left(1 - \frac{h^2\Omega^2}{2m}\right)\mathrm{R}_n + \mathrm{R}'_n\right) \end{bmatrix}.$$

More generally, if we define the components of Ψ as ψ_{ij} and denote $\mu_n = [\mu_{n,1}, \mu_{n,2}]^T$ we can take products of the update equations defined by (7.7) to obtain

$$q_{n+1}^2 = \psi_{11}^2 q_n^2 + \psi_{12}^2 p_n^2 + \mu_{n,1}^2 + 2\psi_{11}\psi_{12} q_n p_n + 2\psi_{11}\mu_{n,1} q_n + 2\psi_{12}\mu_{n,1} p_n, \tag{7.8}$$

$$p_{n+1}^2 = \psi_{21}^2 q_n^2 + \psi_{22}^2 p_n^2 + \mu_{n,2}^2 + 2\psi_{21}\psi_{22} q_n p_n + 2\psi_{21}\mu_{n,2} q_n + 2\psi_{22}\mu_{n,2} p_n, \tag{7.9}$$

$$\begin{aligned} q_{n+1}p_{n+1} = {} & \psi_{11}\psi_{21} q_n^2 + \psi_{12}\psi_{22} p_n^2 + \mu_{n,1}\mu_{n,2} + (\psi_{11}\psi_{22} + \psi_{12}\psi_{21}) q_n p_n \\ & + (\psi_{11}\mu_{n,2} + \psi_{21}\mu_{n,1}) q_n + (\psi_{22}\mu_{n,1} + \psi_{12}\mu_{n,2}) p_n. \end{aligned} \tag{7.10}$$